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MIL-N-24427(SHIPS)
2 June 1971

MILITARY SPECIFICATION

NOZZLES, FIXED, PRE-SET PATTERN, COMBINATION AFFF SOLUTION OR WATER SPRAY

1. SCOPE

1.1 This specification covers nozzles which are for use with Aqueous Film Forming Foam (AFFF) in sea water or fresh water. The nozzles may also be used with plain water in a straight stream or pre-set variable spray pattern.

1.2 Classification. The nozzles shall be of the following sizes and capacities as specified (see 6.2):

Size 3/4-inch - 30 gallons per minute (g.p.m.) at 80 pounds per square inch gage (p.s.i.g.)
Size 1-inch - 60 g.p.m. at 80 p.s.i.g.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
QQ-B-626 - Brass, Leaded and Nonleaded: Rods, Shapes, Forgings, and Flat Products With Finished Edges (Bars and Strip).

MILITARY
MIL-H-775 - Hose, Rubber, Plastic Fabric, or Metal (Including Tubing); and Fittings, Nozzles, and Strainers, Packaging of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-F-1183 - Fittings, Tube, Cast Bronze, Silver-Brazing.
MIL-R-3065 - Rubber.
MIL-F-24059 - Bronze, Nickel Aluminum: Rod, Flat Products with Finished Edges, Shapes and Forgings.
MIL-F-24385 - Fire Extinguishing Agent, Aqueous Film Forming Foam (AFFF) Liquid Concentrate, Six Percent for Fresh and Sea Water.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

FEDERAL
FED-STD-151 - Metals: Test Method.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-810 - Environmental Test Methods.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 4210

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
B-61 - Steam or Valve Bronze Castings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

NATIONAL FIRE PROTECTION ASSOCIATION
Standard 412 - Evaluating Foam Fire Fighting Equipment on Aircraft Rescue and Fire Fighting Venicles.

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Mass. 02110.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production a sample shall be tested as specified in 4.3 (see 6.3).

3.2 Material. Material shall be as specified herein. Nozzles shall be of bronze with rubber O-rings.

3.2.1 Bronze bars. Bronze bars shall conform to MIL-B-24059.

3.2.2 Brass forgings. Brass forgings shall conform with alloy 377 of QQ-B-626.

3.2.3 Cast bronze. Cast bronze shall conform to ASTM B-61.

3.2.4 "O" ring seals. "O"-ring seals shall conform to $SB725BE_3$ of MIL-R-3065.

3.2.5 Connections.

3.2.5.1 Navy ships. Inlet connections for use on board Navy ships shall be of the silver brazing type in accordance with MIL-F-1183, male or female as specified (see 6.2). Female silver brazing connections shall be of the pre-inserted ring type.

3.2.5.2 Land use. For land use, inlet connections shall be National Taper Pipe Threads (NPT) as specified in Handbook H-28, male or female as specified (see 6.2).

3.2.6 Threaded inlet. The body of threaded inlet nozzle shall be equipped with a hex wrenching feature for tightening threads.

3.2.7 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein.

3.3 Design. The nozzles shall be of the air aspirating, variable spray type capable of being pre-set and positively locked in position. The nozzles shall be used with AFFF of MIL-F-24385 in either sea-or fresh-water. Unless otherwise specified (see 6.2), the nozzle shall deliver either a straight stream, or 15- or 30-degree full cone spray pattern. The body shall be indexed for ease of pattern selection. The 1-1/2-inch size nozzle shall have a flow rate of 30 g.p.m. at 80 p.s.i.g. and the 1-inch size nozzle shall have a flow

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rate of 60 g.p.m. at 80 p.s.i.g. Variance from the specified flow rate shall not exceed plus or minus 5 percent when tested in accordance with 4.6.1. The nozzles shall also meet the requirements specified in table I.

Table I - Nozzle characteristics.

Position	Foam Expansion Ratio(min.)	1/4 Drainage Time, Minutes(min.)	Reach ft.
For 3/4-inch nozzle			
Straight stream	4.0	1.2	50
30° cone	3.0	0.7	30
For 1-inch nozzle			
Straight stream	5.5	2.5	65
30° cone	3.0	1.0	50

3.3.1 Dynamic pressure. The nozzles shall withstand the test of 4.6.3. There shall be no leakage of the "O" ring when subjected to the test.

3.3.2 Shock and vibration. The nozzles shall withstand the tests specified in 4.6.6.

3.3.3 High temperature. The nozzles shall withstand the test of 4.6.2 and shall show no leakage.

3.3.4 Salt spray. The nozzles shall operate without malfunction and shall withstand the test of 4.6.7.

3.4 Identification marking. The nozzles shall be identified in accordance with MIL-STD-130.

3.5 Workmanship. All parts and assemblies of the nozzles including castings, forgings, molded parts, stampings, bearings, scales, machine surfaces and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful material. All sharp edges shall be rounded or chamfered.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The supplier shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by the specification. The inspection system shall be in accordance with MIL-I-45208 (see 6.5).

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection.
(b) Quality conformance inspection.
(c) Inspection of preparation for delivery.

4.3 First article inspection. One sample of each size nozzle shall be subjected to first article inspection. First article inspection shall consist of the following examination and tests.

Visual and dimensional	4.5
Discharge capacity	4.6.1
Dynamic pressure	4.6.3
AFFF expansion	4.6.4
Nozzle range	4.6.5
High temperature	4.6.2
Dynamic pressure	4.6.3
Shock	4.6.6
Vibration	4.6.6
Salt spray	4.6.7

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4.4 Quality conformance inspection.

4.4.1 Inspection lot. An inspection lot shall consist of all nozzles of the same size produced under essentially the same conditions and offered for inspection at the same time.

4.4.2 Sampling.

4.4.2.1 Sampling for visual and dimensional examination. A random sample of nozzles shall be selected from each lot in accordance with inspection level S-2 of MIL-STD-105 for the visual and dimensional examination of 4.5. The AQL shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.2.2 Sampling for quality conformance tests. A random sample of nozzles shall be selected from each lot for the test specified in 4.6.2. The AQL shall be 0.65 percent defective.

4.4.3 Quality conformance tests. Each of the sample nozzles selected in accordance with 4.4.2.2 shall be subjected to the test of 4.6.2. Any sample nozzle failing to conform to the tests shall be rejected and if the rejection number exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5 Visual and dimensional examination. Each sample nozzle selected in accordance with 4.4.2.1 shall be visually and dimensionally examined for the major and minor defects specified in table II. Any sample failing to conform to this specification shall be rejected and if the rejection number exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table II - Classification of defects.

Critical	None defined.
Major	
101	Size not as specified.
102	Missing or incorrect parts.
103	Parts damaged so as to interfere with proper operation.
104	Inlet connection not as specified.
105	Dimensions incorrect affecting proper operation.
106	Workmanship not as specified.
Minor	
201	Parts loose or damaged but not interfering with operation.
202	Dimensions incorrect but not interfering with operation.
203	Identification marking missing, incorrect, or illegible.

4.6 Tests procedures.

4.6.1 Discharge capacity. The discharge capacities shall be measured by a flow meter to determine conformance with 3.3.

4.6.2 High temperature. The nozzle shall be subjected to high temperatures in accordance with Method 501, Procedure I of MIL-STD-810. At 160°F. the "O" ring seals shall not leak at 150 p.s.i.g. discharge pressure.

4.6.3 Dynamic pressure. Immediately following the high temperature test, the nozzle shall be subjected to a dynamic pressure of 150 p.s.i.g. to determine conformance with 3.3.1. Any deformation or leakage of the nozzle shall constitute failure of this test.

4.6.4 AFFF expansion. The AFFF analysis shall be conducted in accordance with Standard 412 of the National Fire Protection Association to determine conformance with table I.

4.6.5 Nozzle range. Reach distances shall be determined with the nozzles mounted 1 foot above the ground and angled upward at 10 degrees to determine conformance with table I.

4.6.6 Shock and vibration. The nozzle shall be tested in accordance with grade A, class I of MIL-S-901 for shock and type I of MIL-STD-167 for vibration. Failure to pass these tests shall be cause for rejection.

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4.6.7 Salt spray. The nozzle shall be subjected to salt spray for a period of 5 days in accordance with Method 812 of FED-STD-151 to determine conformance with 3.3.4.

4.7 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be examined and tested as specified in MIL-H-775.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Domestic shipment.

5.1.1 Nozzles.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations, as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment (see 5.3). The requirements and levels of preservation, packaging, packing and marking for shipment shall be as specified by the procuring activity (see 6.2).

5.2.1 The following provide various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.2).

5.2.2 Preservation, packaging, packing, and marking. The nozzles and parts shall be preserved, packaged, packed, and marked in accordance with MIL-H-775. Preservation and packaging shall be level A or C, and packing shall be level A, B, or C, as specified (see 6.2).

5.3 Use of polystyrene (loose-fill) material.

5.3.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.2), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken abard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard or transparent flexible cellular material."

5.3.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Intended use. The nozzles are intended for use in fixed piping systems to generate AFFF for combating class "A" fires (such as wood, paper, textiles, and so forth), class "B" fires (flammable liquid and grease), and for class "C" fires (electrical) up to 250,000 volts at a distance of 10 feet or more. The nozzles are also capable of emitting straight stream or pre-set variable spray patterns.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Sample for first article inspection (see 3.1 and 6.3).
(d) Whether female or male inlet is required (see 3.2.5).
(e) Whether for shipboard or land use (see 3.2.5).
(f) Patterns other than those specified (see 3.3).
(g) Approval for use of polystyrene material (see 5.3).
(h) Levels of preservation, packaging, packing, and marking (see 5.2).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Management control system document. The following management control system document should be included on DD form 1660:

(a) MIL-I-45208 (see 4.1.1)

Preparing activity:
Navy - SH
(Project 4210-N067)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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